

Context: the Boundary layer rolls

- What are rolls ?
- Coherent structure
 - $\lambda = 2-5 \times$ height of PBL
 - Aligned with the mean wind ($\pm 13^\circ$)
 - Due to thermal and dynamical instabilities

- Where are they observed ?
- Cold Air outbreaks (Brümmer and Pohlmann, 2000)
 - Hurricanes (Katsaros et al., 2002)
 - **Windstorms** (Rivière et al., 2020)

- Why interesting ?
- Transfer of heat, momentum, moisture in the Planetary Boundary Layer (Brient et al, 2024)
 - Information on exchanges with the surface (Lfahar et al, 2024)

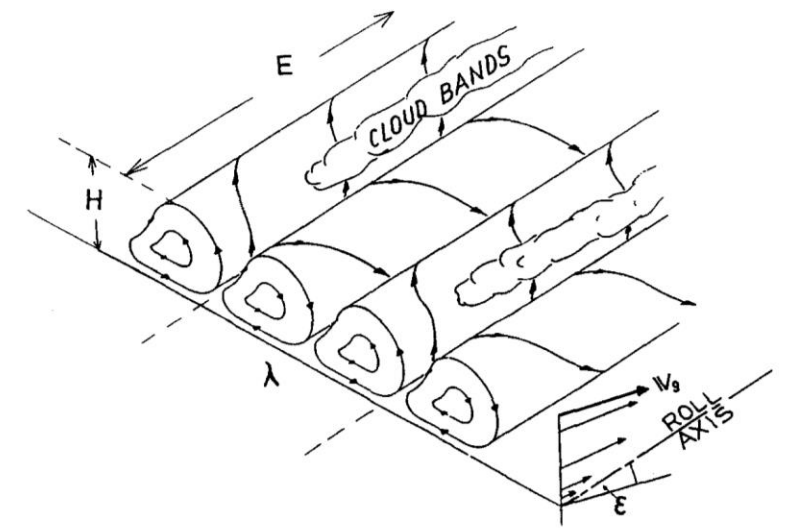
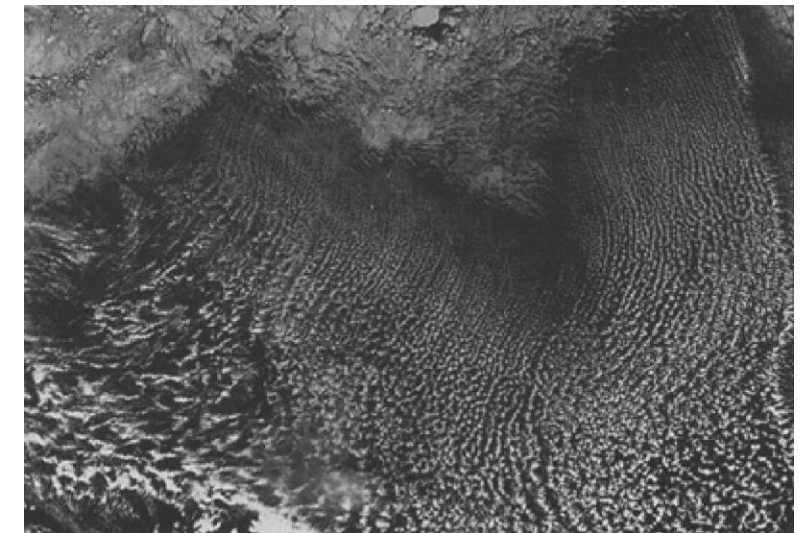


Schéma de structure en rouleaux (Etling et Brown, 1993)



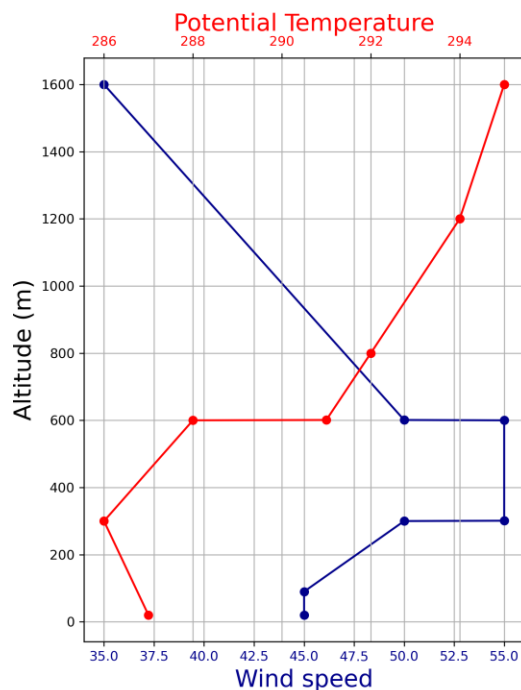
Rouleaux durant épisode de froid au-dessus de la mer de Barents (NOAA 7, AVHRR, 6.4.1988)

Method of study?

- Use of ideal case in Meso-NH in LES
- Vary the parameters (temperature profile + wind profile) to have rolls

Goal ?

- Have a stationnary evolution



Exemple of input profiles

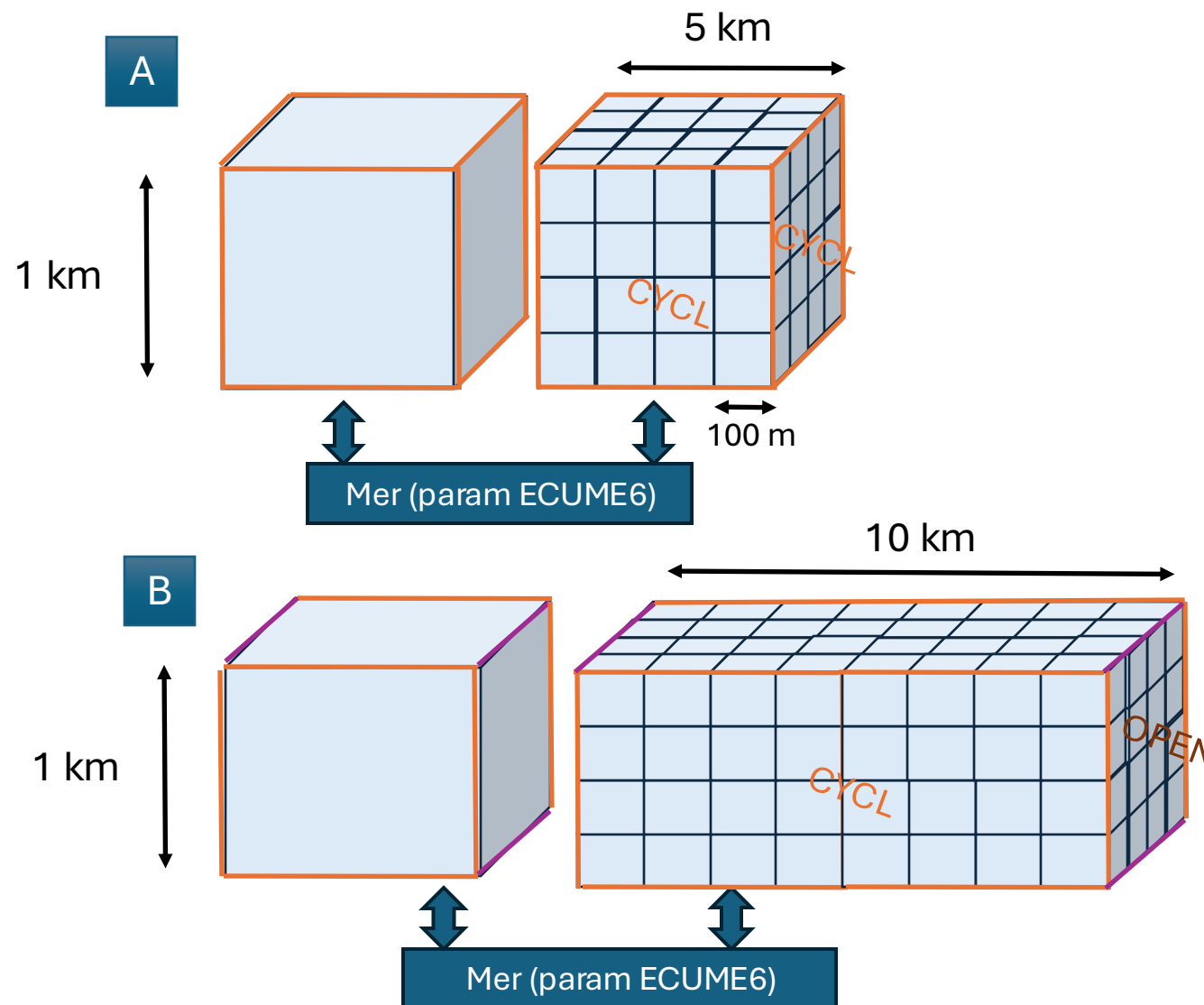


Schéma des différentes expériences